

APPENDIX

SUPPLEMENTARY EXPERIMENTAL RESULTS FOR “WHEN FRONTEND EVENTS LEAK QUERIES: EVENT-AWARE TESTING FOR WEB AUTOCOMPLETE”

Fig. 1 illustrates the distribution of inter-arrival times between consecutive Bing request packets. The distinct temporal patterns and peaks are critical for identifying and differentiating polling-based search activities from background noise.

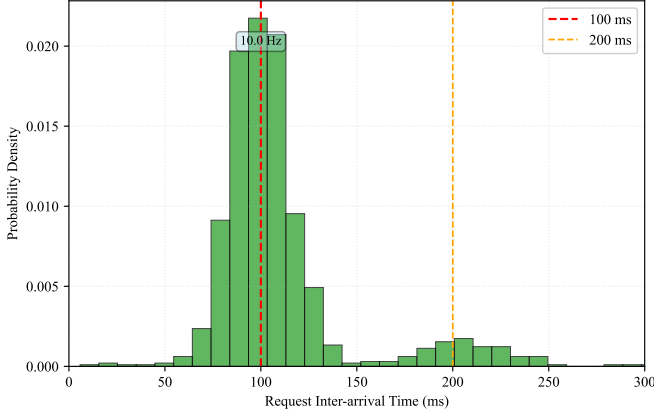


Fig. 1: Statistical Histogram of Bing Request Packet Inter-arrival Times.

Fig. 2 analyzes the sensitivity of the polling model recognition algorithm to time tolerance variations. It demonstrates how the alignment accuracy and classification performance fluctuate as the time tolerance threshold is adjusted, helping to determine the optimal parameter setting.

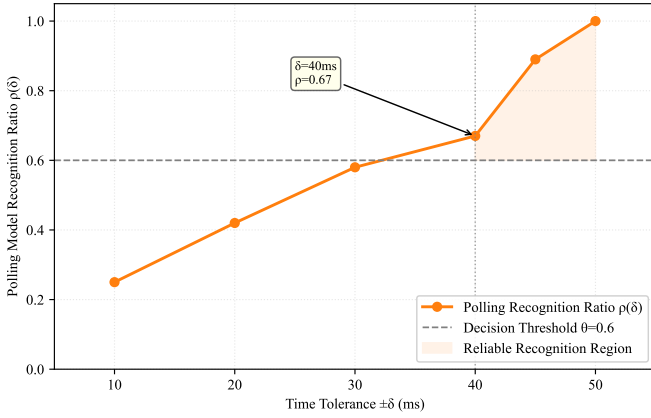


Fig. 2: Time Tolerance Sensitivity Analysis of the Polling Model Recognition Algorithm.

Fig. 3 illustrates the three-layer feature selection system based on the event processing model. It details how universal, callback-specific, and polling-specific features are extracted and filtered to construct the model-adaptive feature set for accurate traffic classification.

Fig. 4 verify the growth pattern of request packet sizes in both callback and polling models. The callback model (Figure 4(a)) exhibits a steady increase in packet size as the keyword length grows, whereas the polling model (Figure 4(b)) shows a distinct size distribution due to its fixed-length polling requests.

Table I summarizes the distribution of packet sizes across different search and e-commerce platforms. It highlights the distinct size intervals characteristic of each platform, which serve as fundamental features for traffic fingerprinting and platform identification.

TABLE I: ASB Traffic Packet Size Distribution by Platform (Key Intervals)

Platform	[0, 64)	[1472, 1536)	[1536, 2000)	[2000, 9000)
Google	40763	18827	0	0
Bing	38564	80418	0	0
Baidu	95351	27082	0	0
Amazon	35147	1941	35	303
eBay	6718	133	35	303
JD	15445	408	86	6152
Taobao	6975	2550	219	8079

Note: Packet size unit is Bytes (B).

Table II presents an ablation study on the position filtering strategy. It demonstrates that incorporating position filtering significantly improves both the feature sequence alignment accuracy and the overall classification performance (MCC) for Bing traffic.

TABLE II: Independent Impact of the Position Filtering Strategy on Bing Feature Sequence Alignment and Classification Performance

Filtering Config.	Bing MCC	Align. Acc.
IP + Size Filtering Only	0.784	71.6%
IP + Size + Position Filtering	0.951	99.8%
No Filtering	0.715	62.3%

Note: Filtering Config. = Filtering Strategy Configuration; Align. Acc. = Feature Alignment Accuracy defined as below.

$$Align.Acc. = \frac{M}{\max(N, \hat{N})} \times 100 \quad (1)$$

where N denotes the ground-truth total number of characters typed by the user; $\hat{N}$ denotes the number of request packets remaining after multi-stage filtering; M denotes the count of perfectly aligned (request packet, character) pairs.

Table III defines the feature combinations used in the controlled experiments. It systematically evaluates the incremental contribution of universal, callback-specific, polling-specific, and model-specific features, along with the Character-based Unified Sequence (CUS), to the classification performance.

Fig. 5 presents the 10-class Matthews Correlation Coefficient (MCC) performance of various classifiers using the model-adaptive feature set across different years (2017-2026).

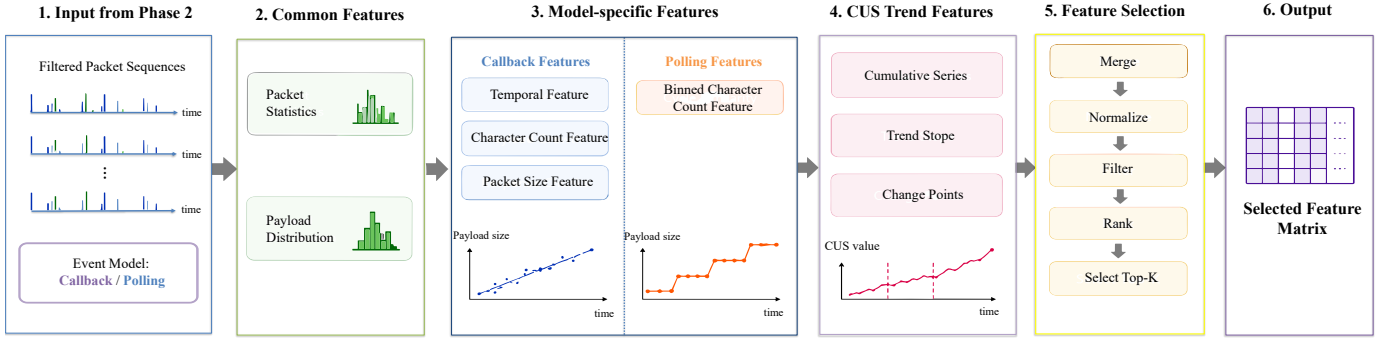
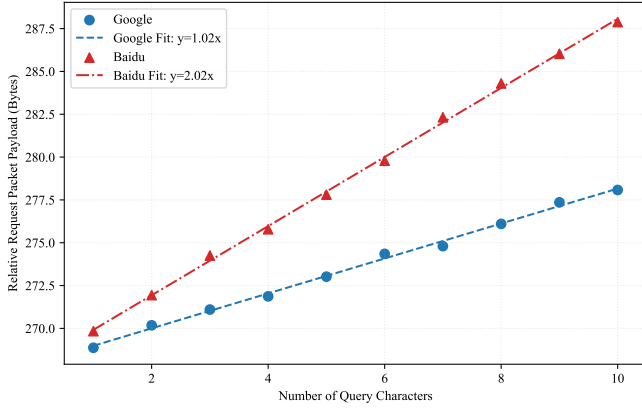
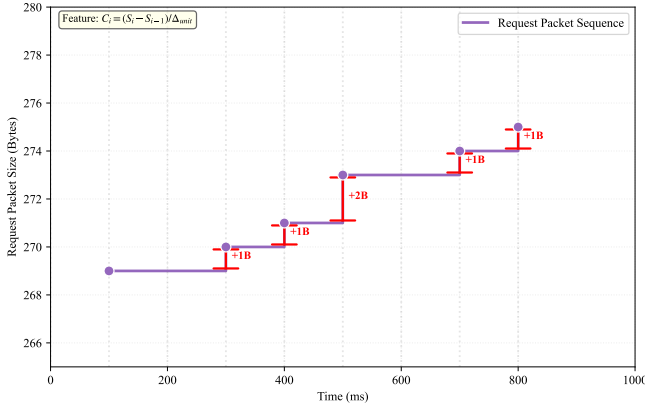


Fig. 3: Feature Selection Based on Event Processing Model.



(a) Callback Model



(b) Polling Model

Fig. 4: Verification of Request Packet Size in Callback Model and Polling Model

TABLE III: Controlled Experiment Settings

Feature Combination	Included Content
F_{base}	Universal packet stat. features only
$F_{base} + F_{callback}$	+ Callback-specific features
$F_{base} + F_{polling}$	+ Polling-specific features
F_{full}	+ Model-specific features + CUS

It demonstrates the long-term stability and robustness of the proposed features against temporal variations in network traffic across Google, Baidu, and Bing.

Table IV compares the average training time (in milliseconds) of different classifiers across the three major search platforms. It provides insights into the computational overhead and training efficiency of traditional machine learning models, deep learning architectures, and large language models.

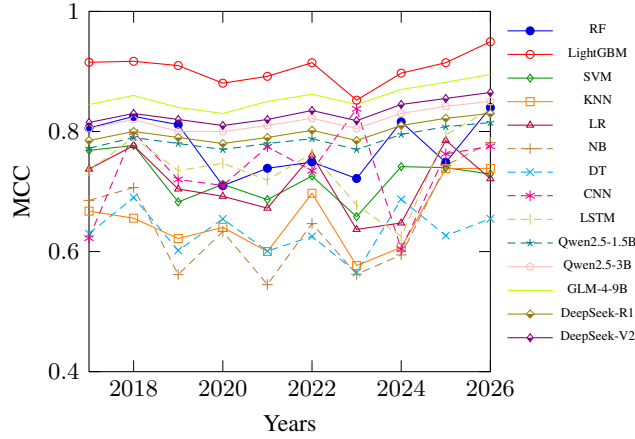
TABLE IV: Average training time of each classifier (ms)

Method	Google	Baidu	Bing
DT	177.79	176.72	184.61
KNN	236.27	371.14	248.71
LR	587.77	429.00	518.67
NB	202.61	216.27	236.33
SVM	106.61	134.59	132.57
RF	1096.99	1054.39	1262.18
Qwen2.5-1.5B	12450.3	12680.1	12520.8
Qwen2.5-3B	18240.5	18650.2	18430.9
GLM-4-9B	34560.2	35820.5	35150.8
DeepSeek-R1-1.5B	13120.8	13450.3	13280.5
DeepSeek-V2-Lite	20540.1	21230.6	20850.4
CNN	5537.46	5771.47	5796.98
LSTM	6643.70	6229.24	6896.99
LightGBM	622.21	612.08	630.68

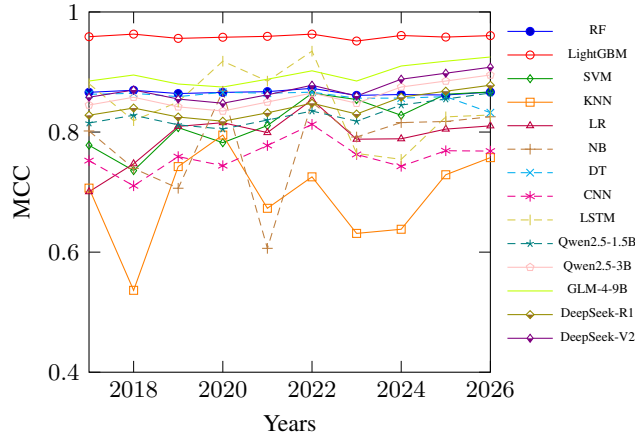
Table V reports the standard deviation of the MCC scores for various side-channel evaluation methods across multiple runs. A lower standard deviation indicates higher stability and reliability of the method's classification performance on single platforms.

Table VI shows the classification performance (MCC) of the proposed method under a real-world manual data collection scenario. It illustrates how the model's accuracy improves as the number of training keywords increases, validating its effectiveness in practical, low-resource settings.

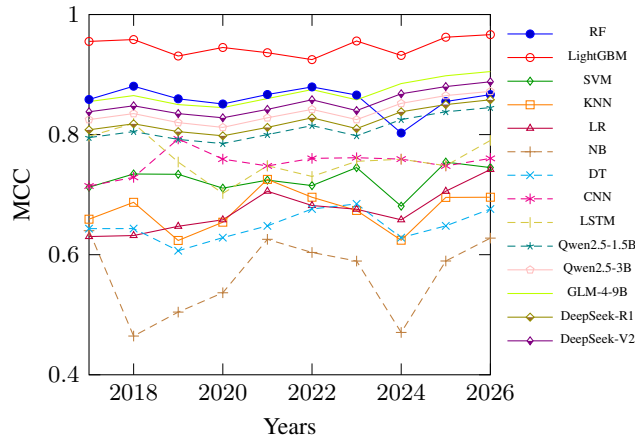
Table VII compares the average number of guessing times required by different side-channel methods to successfully infer keywords on major e-commerce platforms. The proposed



(a) Google



(b) Baidu



(c) Bing

Fig. 5: 10-Class MCC of Model-Adaptive Feature Set Across Different Years

TABLE V: Standard Deviation of MCC for Each Method on Single Platforms

Method	Google	Baidu	Bing
Chapman [1]	0.034	0.029	0.041
Lescisin [2]	0.027	0.031	0.039
Oh [3]	0.022	0.025	0.033
Monaco [4]	0.051	0.048	0.055
FEAST	0.012	0.015	0.018

FEAST method requires significantly fewer attempts, demonstrating superior efficiency in real-world attack scenarios.

Table VIII details the Top-1, Top-3, and Top-5 keyword guessing accuracies of various side-channel methods across four major e-commerce platforms. It comprehensively demonstrates that the proposed method achieves the highest accuracy across all evaluation metrics and platforms.

Table IX presents the statistical distribution of Bing interference packet payload size.

TABLE VI: Results of Real Manual Data Collection Experiment (MCC with Different Numbers of Training Keywords)

Platform	Number of Training Keywords									
	10	20	30	40	50	60	70	80	90	100
Google	0.3125	0.3873	0.4144	0.4384	0.4871	0.5143	0.5344	0.5681	0.5912	0.6059
Baidu	0.3381	0.4127	0.4593	0.4945	0.5122	0.5596	0.5911	0.6163	0.6344	0.6437
Bing	0.3817	0.4033	0.4511	0.5013	0.5387	0.5652	0.5865	0.6024	0.6122	0.6145

TABLE VII: Comparison of Keyword Guessing Times on Major E-commerce Platforms

Method	Amazon	eBay	Taobao	JD.com	Avg. Times
Chapman [1]	9.67	9.41	7.73	8.12	8.73
Lescisin [2]	8.91	7.72	6.34	7.58	7.64
Oh [3]	6.78	6.61	5.21	5.47	6.02
Monaco [4]	13.38	12.12	10.47	11.91	11.97
FEAST	4.87	3.99	3.08	3.94	3.97

Note: Evaluated on four major platforms. Avg. Times = Average Guessing Times.

TABLE VIII: Comparison of Keyword Guessing Accuracy of Different Side-channel Evaluation Methods on Four Major E-commerce Platforms

Method	Platform	Top-1 Acc	Top-3 Acc	Top-5 Acc
Chapman [1]	Amazon	0.276	0.441	0.518
	eBay	0.284	0.452	0.529
	Taobao	0.312	0.487	0.561
	JD.com	0.298	0.463	0.542
	Average	0.293	0.461	0.538
Lescisin [2]	Amazon	0.389	0.571	0.657
	eBay	0.396	0.583	0.669
	Taobao	0.421	0.612	0.698
	JD.com	0.407	0.594	0.681
	Average	0.403	0.590	0.676
Oh [3]	Amazon	0.451	0.631	0.712
	eBay	0.462	0.643	0.724
	Taobao	0.487	0.671	0.752
	JD.com	0.469	0.653	0.734
	Average	0.467	0.650	0.731
Monaco [4]	Amazon	0.161	0.276	0.348
	eBay	0.168	0.285	0.358
	Taobao	0.187	0.312	0.389
	JD.com	0.173	0.294	0.367
	Average	0.172	0.292	0.366
FEAST	Amazon	0.593	0.737	0.789
	eBay	0.604	0.746	0.797
	Taobao	0.621	0.763	0.812
	JD.com	0.608	0.751	0.801
	Average	0.607	0.749	0.800

TABLE IX: Statistics of Bing Interference Packet Payload Sizes

196	197	198	199	200	201	202	203	204	205
1	11	77	96	96	90	81	54	32	6

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- [1] P. Chapman and D. Evans, "Automated black-box detection of side channel vulnerabilities in web applications," in ACM Conference on Computer & Communications Security, 2011, p. 263.
- [2] M. Lescisin, K. Eykholt, Q. Mahmoud, I. Evtimov, S. Narayan, E. Fernandes, F. Brown, B. Li, D. Stefan, and A. Rahmati, "Tools for active and passive network side-channel detection for web applications," in 12th USENIX Workshop on Offensive Technologies (WOOT 18), 2018.
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- [4] J. V. Monaco, "What are you searching for? A remote keylogging attack on search engine autocomplete," in USENIX Security Symposium, 2019.